Ajwad Tazrif Hussyne

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Profile

Electrical Engineering student with hands-on experience in electrical testing, instrumentation, troubleshooting, embedded systems, and power system design. Experienced with oscilloscopes, measurement systems, FPGA hardware, and engineering documentation. Motivated to gain practical experience in commissioning, maintenance, power systems testing, and electrical field services.

Education

University of Calgary, Schulich School of Engineering — *Expected April 2028*
Bachelor of Science in Electrical Engineering

Relevant Coursework: Power Systems, Instrumentation Systems, Control Systems, Analog and Digital Circuits, Embedded Systems, Signals and Systems, Engineering Design

Core Skills

Electrical Testing & Troubleshooting: oscilloscopes, electrical measurements, instrumentation, signal monitoring, troubleshooting, root cause analysis, hardware validation

Power Systems & Engineering: grid-connected PV systems, inverter sizing, power system fundamentals, electrical calculations, system monitoring, engineering documentation

Technical Tools: MATLAB, SystemVerilog, C/C++, Xilinx Vivado, LTSpice, NI Multisim, PVsyst, AutoCAD, Fusion 360, Microsoft Excel, Word, PowerPoint

Field & Professional Skills: QA/QC mindset, safety awareness, technical reporting, teamwork, communication, adaptability, preventative maintenance mindset

Relevant Experience

Program Assistant — *June 2025 – July 2025*
SHAD Canada – University of Calgary

- Supported hands-on engineering workshops involving equipment setup, troubleshooting, monitoring, and technical coordination.
- Assisted with technical activities, documentation, and operational logistics in fast-paced team environments.

Math Instructor — *Jan 2024 – Oct 2024*
Mathnasium

- Developed analytical, problem-solving, and communication skills through technical instruction and student performance analysis.

Engineering Projects

Dual-Axis Solar Tracker Using LDR Sensors and Servo Motors — *Jan 2026 – Apr 2026*
Embedded Systems and Renewable Energy Project

- Developed Arduino-based solar tracking system integrating sensors, servo motors, embedded control logic, and electrical wiring.
- Conducted repeated testing, troubleshooting, calibration, and performance validation under varying operating conditions.
- Assisted with mechanical integration, system assembly, and engineering documentation using Fusion 360 and technical reports.

Grid-Connected Solar PV System Design (6.3 kWp) — *Sep 2025 – Dec 2025*
Power Systems Design Project

- Designed and simulated residential grid-connected PV system using PVsyst and engineering calculations.
- Evaluated inverter sizing, module configuration, and system performance to support technical decision-making.

Discrete ADC Design Using FPGA (PWM, R2R DAC, SAR Logic) — *Sep 2025 – Dec 2025*
Instrumentation and Hardware Testing Project

- Designed and implemented FPGA-based analog-to-digital conversion system integrating PWM generation, comparator interfacing, and SAR logic.
- Performed electrical testing, signal validation, troubleshooting, and waveform analysis using oscilloscopes and measurement equipment.
- Investigated signal behavior, corrected hardware integration issues, and documented testing procedures and engineering results.

Embedded Control System with Obstacle Detection — *Jan 2025 – Apr 2025*
Embedded Systems Project

- Integrated ultrasonic sensors, embedded controllers, and actuator logic within an electromechanical system.
- Conducted system troubleshooting, signal verification, and corrective debugging to improve operational reliability.

Extra-Curricular Involvement

AASEE Finance Team — *Sep 2025 – Present*
University of Calgary

- Managed budgeting records, sponsorship applications, expense tracking including outreach planning, and financial coordination for STEM outreach activities and events.

Volunteering Experience

STEM Learning Facilitator — *Apr 2025 – Present*
AASEE

- Delivered hands-on STEM workshops and engineering demonstrations to 1500+ students across schools and outreach events.
- Assisted with technical setup, troubleshooting, and activity coordination during engineering workshops.

Additional Information

- Strong interest in electrical load listing, preventative maintenance, protection systems, and power system field services.
- Comfortable working in hands-on technical and industrial environments with field-style responsibilities.
- Quick learner with strong curiosity toward electrical equipment testing, system reliability, and utility operations.